EQUIVALENCE CRITERION FOR A QUOTIENT-STACK MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-groupoids.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 046Q: sources/stacks-chow/tags.txt, source lines 5481–5481. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 046S: sources/stacks-chow/tags.txt, source lines 5483–5483. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 046N: sources/stacks-chow/tags.txt, source lines 5478–5478. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 044X: sources/stacks-chow/tags.txt, source lines 5416–5416. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0469: sources/stacks-chow/tags.txt, source lines 5464–5464. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumption as in Lemma 046Q. The morphism of quotient stacks*

$$[f] : [U/R] \longrightarrow [U'/R']$$

is an equivalence if and only if

(1) (U, R, s, t, c) *is the restriction of* (U', R', s', t', c') *via* $f : U \to U'$*, and*
(2) *the map*

$$U \times_{f,U',t'} R' \xrightarrow{pr_1} R' \xrightarrow[s']{} U' \quad \text{with} \quad h : U \times_{f,U',t'} R' \to U'$$

is a surjection of sheaves.

Part (2) holds for example if $\{h : U \times_{f,U',t'} R' \to U'\}$ *is an fppf covering, or if* $f : U \to U'$ *is a surjection of sheaves, or if* $\{f : U \to U'\}$ *is an fppf covering.*

Proof. We already know that part (1) is equivalent to fully faithfulness by Lemma 046S. Hence we may assume that (1) holds and that $[f]$ is fully faithful. Our goal is to show, under these assumptions, that $[f]$ is an equivalence if and only if (2) holds. We may use Stacks, Lemma 046N which characterizes equivalences.

Date: stacks-project:spaces-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Assume (2). We will use Stacks, Lemma 046N to prove $[f]$ is an equivalence. Suppose that T is a scheme and $x' \in \mathrm{Ob}([U'/R']_T)$. There exists a covering $\{g_i : T_i \to T\}$ such that $g_i^* x'$ is the image of some element $a'_i \in U'(T_i)$, see Lemma 044X. Hence we may assume that x' is the image of $a' \in U'(T)$. By the assumption that h is a surjection of sheaves, we can find an fppf covering $\{\varphi_i : T_i \to T\}$ and morphisms $b_i : T_i \to U \times_{g,U',t'} R'$ such that $a' \circ \varphi_i = h \circ b_i$. Denote $a_i = \mathrm{pr}_0 \circ b_i : T_i \to U$. Then we see that $a_i \in U(T_i)$ maps to $f \circ a_i \in U'(T_i)$ and that $f \circ a_i \cong_{T_i} h \circ b_i = a' \circ \varphi_i$, where $\cong_{T_i}$ denotes isomorphism in the fibre category $[U'/R']_{T_i}$. Namely, the element of $R'(T_i)$ giving the isomorphism is $\mathrm{pr}_1 \circ b_i$. This means that the restriction of x to T_i is in the essential image of the functor $[U/R]_{T_i} \to [U'/R']_{T_i}$ as desired.

Assume $[f]$ is an equivalence. Let $\xi' \in [U'/R']_{U'}$ denote the object corresponding to the identity morphism of U'. Applying Stacks, Lemma 046N we see there exists an fppf covering $\mathcal{U}' = \{g'_i : U'_i \to U'\}$ such that $(g'_i)^* \xi' \cong [f](\xi_i)$ for some ξ_i in $[U/R]_{U'_i}$. After refining the covering $\mathcal{U}'$ (using Lemma 044X) we may assume ξ_i comes from a morphism $a_i : U'_i \to U$. The fact that $[f](\xi_i) \cong (g'_i)^* \xi'$ means that, after possibly refining the covering $\mathcal{U}'$ once more, there exist morphisms $r'_i : U'_i \to R'$ with $t' \circ r'_i = f \circ a_i$ and $s' \circ r'_i = \mathrm{id}_{U'} \circ g'_i$. Picture

$$\begin{array}{ccccc} U & \xleftarrow{a_i} & U'_i & & \\ \downarrow{\scriptstyle f} & \swarrow{\scriptstyle r'_i} & \downarrow{\scriptstyle g'_i} & & \\ U' & \xleftarrow{t'} & R' & \xrightarrow{s'} & U' \end{array}$$

Thus $(a_i, r'_i) : U'_i \to U \times_{g,U',t'} R'$ are morphisms such that $h \circ (a_i, r'_i) = g'_i$ and we conclude that $\{h : U \times_{g,U',t'} R' \to U'\}$ can be refined by the fppf covering $\mathcal{U}'$ which means that h induces a surjection of sheaves, see Topologies on Spaces, Lemma 0469.

If $\{h\}$ is an fppf covering, then it induces a surjection of sheaves, see Topologies on Spaces, Lemma 0469. If $U' \to U$ is surjective, then also h is surjective as s has a section (namely the neutral element e of the groupoid in algebraic spaces). $\square$